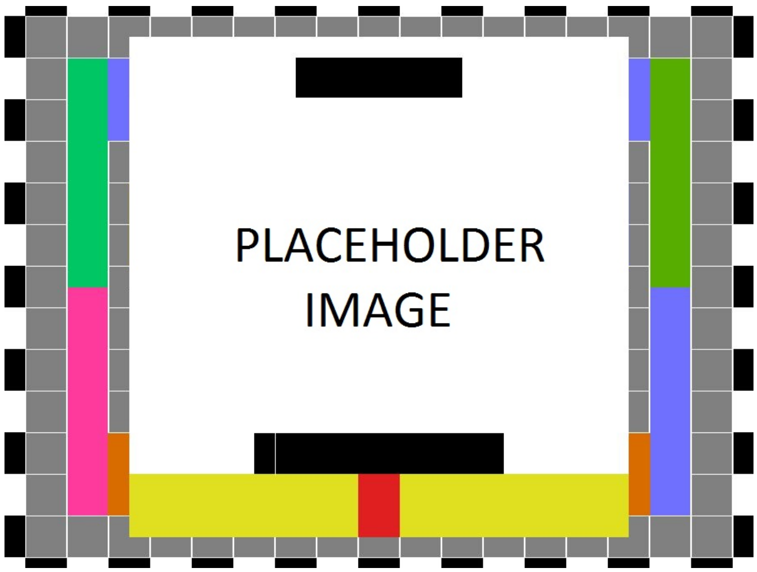
Department of Computer Science & Engineering
The University of Texas at Arlington

System Requirements Specification
CSE 4316: Senior Design I
Fall 2021



Team Name
Product Name

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Revision History

Revision	Date	Author(s)	Description
0.1	10.01.2015	GH	document creation
0.2	10.05.2015	AT, GH	complete draft
0.3	10.12.2015	AT, GH	release candidate 1
1.0	10.20.2015	AT, GH, CB	official release
1.1	10.31.2015	AL	added customer change requests

Contents

LIST OF FIGURES

1 Product Concept

This section provides a high-level statement of your product concept - what it is intended to do and how it is intended to be used. Include in this header paragraph, a brief synopsis of what is described here. For example, this header paragraph might say something like: "This section describes the purpose, use and intended user audience for the X product. X is a system that performs Y. Users of X will be able to Z..."

1.1 Purpose and Use

This is where you describe in a brief, yet clear and concise, manner what your product should do and how you expect it should be used.

1.2 Intended Audience

This is where you describe the intended audience(s) of your product. If this product were to be made available publicly or commercially, who would purchase or use it? Is the product designed for a particular customer, or an overall class of customers? Is it intended for general use, or is it a specific component of a more complex system?

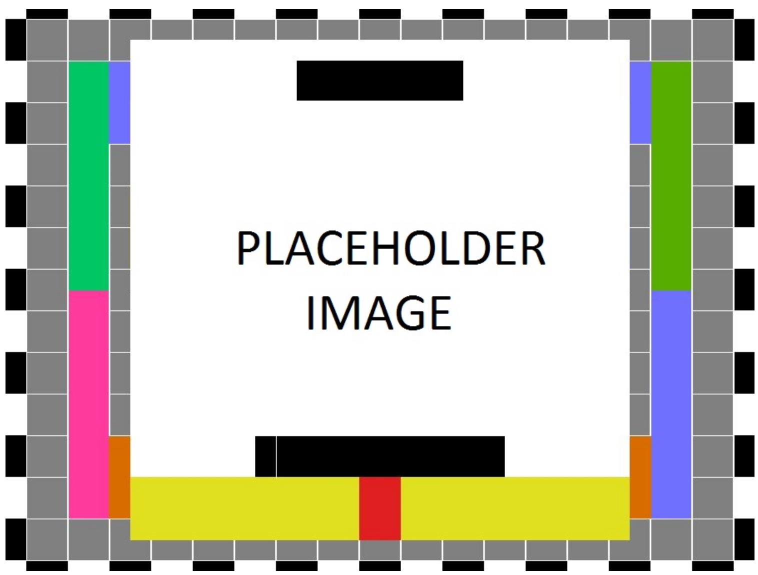

Figure 1: X conceptual drawing

2 PRODUCT DESCRIPTION

3 PRODUCT DESCRIPTION

This section provides an overview of the smart plant monitoring system, including its primary features, functions, and user interactions. The system is designed to monitor plant health using an integrated camera and environmental sensors, and to provide actionable insights through a user interface. From the perspective of end users, the system enables real-time monitoring and feedback. For maintainers and developers, it provides a structured backend for data processing and analysis. The key components, system behavior, and user interactions are described below.

3.1 FEATURES & FUNCTIONS

The smart plant monitoring system is designed to capture plant images and environmental data, process this information, and provide users with plant health insights.

The system consists of the following core components:

- **Camera Module:** Captures images of the plant at regular intervals.
- **Processing Unit:** Processes captured data and determines plant health status using backend logic.
- **Backend Server:** Stores data and performs analysis.
- **User Interface (Dashboard):** Displays plant health status, historical data, and alerts.

The system performs the following functions:

- Captures plant images automatically
- Processes data to determine plant health status
- Stores data for historical tracking
- Displays results and alerts to the user

The system does **not** perform the following:

- It does not physically interact with or modify the plant (no watering or actuation)
- It does not provide medical or agricultural guarantees
- It does not operate outdoors in extreme environmental conditions

External elements include the Internet (for data transfer) and optional cloud storage services.

3.2 EXTERNAL INPUTS & OUTPUTS

The system receives input from both users and hardware components, and produces outputs for user interpretation.

Name	**Description**	**Use**
Camera Image	Image of plant captured periodically	Used for plant health analysis
User Input	User settings (e.g., thresholds)	Configures system behavior
Health Status	Computed plant condition (e.g., healthy/unhealthy)	Displayed to user
Alerts	Notifications for abnormal conditions	Warns user of issues
Historical Data	Stored past readings and images	Used for tracking trends

3.3 PRODUCT INTERFACES

The system provides a user-facing dashboard interface and internal system interfaces.

User Interface:

- Displays plant health status (e.g., healthy, warning, critical)
- Provides historical graphs of plant conditions
- Displays alerts and notifications

Hardware Interface:

- Camera connected to processing unit
- Sensors connected via microcontroller (e.g., Arduino or Raspberry Pi)

Software Interface:

- Backend server processes incoming data
- Communication via APIs between hardware and backend
- Optional cloud database for storage

4 Customer Requirements

Include a header paragraph specific to your product here. Customer requirements are those required features and functions specified for and by the intended audience for this product. This section establishes, clearly and concisely, the "look and feel" of the product, what each potential end-user should expect the product do and/or not do. Each requirement specified in this section is associated with a specific customer need that will be satisfied. In general Customer Requirements are the directly observable features and functions of the product that will be encountered by its users. Requirements specified in this section are created with, and must not be changed without, specific agreement of the intended customer/user/sponsor.

4.1 Requirement Name

4.1.1 Description

A detailed description of the feature/function that satisfies the requirement. For example: *The GUI background will be slate blue. This specific color is required in order to ensure that the GUI matches other similar software products offered by the customer. Slate blue is specified as #007FFF, using six-digit hexadecimal color specification.* It is acceptable and advisable to include drawings/graphics in the description if it aids understanding of the requirement.

4.1.2 Source

The source of the requirement (e.g. customer, sponsor, specified team member (by name), federal regulation, local laws, CSE Senior Design project specifications, etc.)

4.1.3 Constraints

A detailed description of realistic constraints relevant to this requirement. Economic, environmental, social, political, ethical, health & safety, manufacturability, and sustainability should be discussed as appropriate.

4.1.4 Standards

A detailed description of any specific standards that apply to this requirement (e.g. *NSTM standard xx.xxx.x. color specifications [1].* Standards exist for practically everything (ATC standard fuses, IEEE 802.15.4 embedded wireless, TLS 1.3 encryption, etc.), so be sure that you research and document which ones will be followed in meeting this requirement.

4.1.5 Priority

The priority of this requirement relative to other specified requirements. Use the following priorities:

- Critical (must have or product is a failure)
- High (very important to customer acceptance, desirability)
- Moderate (should have for proper product functionality);
- Low (nice to have, will include if time/resource permits)
- Future (not feasible in this version of the product, but should be considered for a future release).

4.2 Requirement Name

4.2.1 Description

Detailed requirement description...

4.2.2 SOURCE

Source

4.2.3 CONSTRAINTS

Detailed description of applicable constraints...

4.2.4 STANDARDS

List of applicable standards

4.2.5 PRIORITY

Priority

5 Packaging Requirements

The finished prodect will include an overhead light fixture with the camera rig installed, a web client accessable on any device with a web browser, and access to AI image processing via a cloud server. Packaging for the camera rig will require significant assembly time. Since the light fixture is designed to mount to any pre-exisiting shelf, the user must be able to adjust the mounting points of the rig to attach it to their shelf.

5.1 Hardware Packaging

5.1.1 Description

Hardware for the camera system will be partally assembled, but mounting the light fixture will require adjustment by the user. The hardware package will include custom mounting hardware for their shelf.

5.1.2 Source

Project Charter

5.1.3 Constraints

Size of packaging, and complexity of assembly will restrict the amount of neccessary pre assembled parts.

5.1.4 Standards

None

5.1.5 Priority

Medium

5.2 Software Packaging

5.2.1 Description

Access to the main control interface for the plant detection system will be through a web server accessable through a web browser. All AI image processing is done on the cloud, and all image capture is done on an included computer. No additional software should be required.

5.2.2 Source

Project Charter

5.2.3 Constraints

Processing power of the onboard computer limits the amount of image processing done on it, as well as its ability to host a web server.

5.2.4 Standards

None

5.2.5 Priority

Medium

6 Performance Requirements

Preformance requirements for the plant monitor include are reliability of the system, and the speed of scanning, and image processing

6.1 Reliability

6.1.1 Description

Reliabilty refers to the both image capture and AI detection. The system for the image capture must be able to take consistant pictures of leaves with acceptable quality, lighting, and angle. The AI algorithm must detect any signs of disease with high accuracy without triggering false alarms. Every step of image capture and processing must be done without human intervention.

6.1.2 Source

Project Charter

6.1.3 Constraints

Constraints include image quality, lighting quality, and angle of leaves relative to the camera.

6.1.4 Standards

None

6.1.5 Priority

High

6.2 Speed

6.2.1 Description

Speed refers to the time it take from initializing image capture, to a result sent to the web client. A low time between initialize and result provides a smoother experience for the user and less time when they are unable to physically interact with the plant rack.

6.2.2 Source

Project Charter

6.2.3 Constraints

Constraints include processing speed, network speeds, and time for image capturing by the camera rig.

6.2.4 Standards

None

6.2.5 Priority

Low

7 Safety Requirements

This section defines the safety requirements for the smart plant monitoring system. The system includes a camera mounted on a stand, sensors, and backend processing components. Safety considerations focus on preventing electrical hazards, ensuring safe physical design, and protecting users from environmental or operational risks. The product must be safe for indoor use, especially in home or laboratory environments, and must not expose users to harmful materials, electrical shocks, or unsafe mechanical structures.

7.1 Laboratory equipment lockout/tagout (LOTO) procedures

7.1.1 Description

Any fabrication equipment provided used in the development of the project shall be used in accordance with OSHA standard LOTO procedures. Locks and tags are installed on all equipment items that present use hazards, and ONLY the course instructor or designated teaching assistants may remove a lock. All locks will be immediately replaced once the equipment is no longer in use.

7.1.2 Source

CSE Senior Design laboratory policy

7.1.3 Constraints

Equipment usage, due to lock removal policies, will be limited to availability of the course instructor and designed teaching assistants.

7.1.4 Standards

Occupational Safety and Health Standards 1910.147 - The control of hazardous energy (lockout/tagout).

7.1.5 Priority

Critical

7.2 National Electric Code (NEC) wiring compliance

7.2.1 Description

Any electrical wiring must be completed in compliance with all requirements specified in the National Electric Code. This includes wire runs, insulation, grounding, enclosures, over-current protection, and all other specifications.

7.2.2 Source

CSE Senior Design laboratory policy

7.2.3 Constraints

High voltage power sources, as defined in NFPA 70, will be avoided as much as possible in order to minimize potential hazards.

7.2.4 Standards

NFPA 70

7.2.5 Priority

Critical

7.3 RIA robotic manipulator safety standards

7.3.1 Description

Robotic manipulators, if used, will either housed in a compliant lockout cell with all required safety interlocks, or certified as a "collaborative" unit from the manufacturer.

7.3.2 Source

CSE Senior Design laboratory policy

7.3.3 Constraints

Collaborative robotic manipulators will be preferred over non-collaborative units in order to minimize potential hazards. Sourcing and use of any required safety interlock mechanisms will be the responsibility of the engineering team.

7.3.4 Standards

ANSI/RIA R15.06-2012 American National Standard for Industrial Robots and Robot Systems, RIA TR15.606-2016 Collaborative Robots

7.3.5 Priority

Critical

8 Security Requirements

Include a header paragraph specific to your product here. In this section specify any requirements related to information security or privacy. This may include items such as encryption standards, data storage procedures, authentication, password strength, etc.

8.1 Requirement Name

8.1.1 Description

Detailed requirement description...

8.1.2 Source

Source

8.1.3 Constraints

Detailed description of applicable constraints...

8.1.4 Standards

List of applicable standards

8.1.5 Priority

Priority

9 Maintenance & Support Requirements

Include a header paragraph specific to your product here. Maintenance and support requirements address items specific to the ongoing maintenance and support of your product after delivery. Think of these requirements as if you were the ones who would be responsible for caring for customers/end user after the product is delivered in its final form and in use "in the field". What would you require to do this job? Specify items such as: where, how and who must be able to maintain the product to correct errors, hardware failures, etc.; required support/troubleshooting manuals/guides; availability/documentation of source code; related technical documentation that must be available for maintainers; specific/unique tools required for maintenance; specific software/environment required for maintenance; etc.

9.1 Requirement Name

9.1.1 Description

Detailed requirement description...

9.1.2 Source

Source

9.1.3 Constraints

Detailed description of applicable constraints...

9.1.4 Standards

List of applicable standards

9.1.5 Priority

Priority

10 Other Requirements

Include a header paragraph specific to your product here. In this section specify anything else that is required for the product to be deemed complete. Include requirements related to customer setup and configuration if not specified in a previous requirement. Add any known requirements related to product architecture/design, such as modularity, extensibility (for future enhancements), or adaptation for a specific programming language. Consider requirements such as portability of your source code to various platforms (Windows, Linux, Unix Mac OS, etc.).

10.1 Requirement Name

10.1.1 Description

Detailed requirement description...

10.1.2 Source

Source

10.1.3 Constraints

Detailed description of applicable constraints...

10.1.4 Standards

List of applicable standards

10.1.5 Priority

Priority

11 Future Items

12 Future Items

This section summarizes features and functionalities that were considered during the design of the smart plant monitoring system but will not be implemented in the current prototype due to limitations in time, resources, and technical complexity. These features may be included in future versions of the system.

12.1 Mobile application support

12.1.1 Description

A dedicated mobile application will allow users to monitor plant health, receive notifications, and control system settings remotely.

12.1.2 Source

User experience enhancement goals

12.1.3 Constraints

Requires additional development time, cross-platform compatibility, and backend integration.

12.1.4 Standards

Mobile application development standards (iOS/Android guidelines)

12.1.5 Priority

Future

12.2 Advanced AI-based plant disease detection

12.2.1 Description

The system will use machine learning models to detect specific plant diseases from captured images and provide detailed diagnostics and recommendations.

12.2.2 Source

Project design discussion

12.2.3 Constraints

Requires large datasets, model training, and higher computational resources.

12.2.4 Standards

AI/ML best practices and data handling guidelines

12.2.5 Priority

Future

12.3 Cloud-based data analytics and sharing

12.3.1 Description

The system will support cloud integration for storing plant data, enabling long-term analytics and sharing data across multiple devices.

12.3.2 Source

System scalability considerations

12.3.3 CONSTRAINTS

Requires cloud infrastructure, security implementation, and ongoing maintenance.

12.3.4 STANDARDS

Cloud computing and data security standards

12.3.5 PRIORITY

Future

REFERENCES

[1] K. S. Rubin, *Essential Scrum: A Practical Guide to the Most Popular Agile Process*, 1st ed. Addison-Wesley Professional, 2012.